WEAPONS AND TECHNOLOGY

## CONGREVE ROCKETS

The British discovered the effectiveness of rockets in war when the weapon was used against them by the army of Mysore in India in the late 18th century. Working at the Royal Arsenal at Woolwich, William Congreve developed the type that bears his name between 1801 and 1805. Propelled by black powder, Congreve rockets had a range of up to 2 miles (3 km), although accuracy was poor and the warhead frequently exploded prematurely. The rockets could be fitted with a variety of warheads, including shrapnel and incendiary, ranging in size from 3 to 24 pounds. They were used by the Royal Navy in the Napoleonic Wars, notably at Copenhagen in 1807, and in the bombardment of Baltimore during the 1812 war with the United States—as immortalized in the line of the US national anthem, "And the rockets' red glare, the bombs bursting in air." Congreve's rockets were employed as weapons during the First Opium War, fired by British gunboats to destroy Chinese junks.



**Self-propelled missiles**
The Congreve rocket was improved in 1815 with a guide stick fitted to the base of the rocket head, rather than along the side, reducing drag.



WAR OF 1812

## BOMBARDMENT OF BALTIMORE



**Date** September 13–14, 1814
**Forces** British: 19 ships; Americans: none
**Losses** British: none; Americans: none

**Location** Baltimore, Maryland

After capturing and burning Washington in August 1814, the British chose Baltimore as their next target. The city was a shipbuilding center and the home port of privateers raiding British merchant shipping. The Americans had no naval force adequate to resist Vice-Admiral Sir Alexander Cochrane's North American Station fleet, but the entrance to the harbor was covered by the guns of Fort McHenry and blocked by a line of sunken merchant ships.

The British put soldiers ashore to attack the city by land, but this force halted in the face of stiff resistance. At 6:30 a.m. on September 13 Cochrane began the naval attack. Anchored out of range of the fort's guns, four bomb vessels firing mortars kept up a sustained bombardment, while the rocket ship *Erebus* fired its Congreve rockets through a porthole from below deck. In the afternoon the ships shifted closer inshore and came under heavy fire from Fort McHenry. Cochrane eventually had *Erebus* towed back out of range. In foul weather the bombardment continued through the night, creating a spectacular pyrotechnic display, but to little practical effect.

At dawn Lieutenant Colonel George Armistead had a large stars-and-stripes flag raised over the fort, inspiring the lawyer and poet Francis Scott Key to write the verses that would become the American national anthem. The British gave up the attempt to take Baltimore.

WITNESS TO WAR

### LIEUTENANT COLONEL GEORGE ARMISTEAD

COMMANDER OF THE GARRISON AT FORT MCHENRY

BOMBARDMENT OF BALTIMORE

*"One of the 24 pounders on the south-west bastion, under the immediate command of Capt. Nicholson, was dismounted by a shell, the* ***explosion from which … wounded several of his men****; the bustle produced in removing the wounded and remounting the gun, probably induced the enemy to suspect we were in a* ***state of confusion****, as he* ***brought in three bomb-ships*** *to what I believed to be good striking distance. I immediately* ***ordered a fire to be opened*** *… within half an hour those intruders again sheltered themselves by* ***withdrawing beyond our reach****. We gave three cheers and again ceased firing."*

SECOND BARBARY WAR

## ACTION OF 17 JUNE



**Date** June 17, 1815
**Forces** Americans: 3 frigates, 6 other ships; Algerians: 1 frigate
**Losses** Americans: none; Algerians: none

**Location** Western Mediterranean

In March 1815 the US Congress authorized the dispatch of a naval force to North Africa, where the Barbary pirates had resumed attacks on American shipping and the capture and ransom of American sailors. Commodore Stephan Decatur reached Gibraltar with a squadron of nine ships in mid-June. Hearing that Algerian raiders were at sea, he immediately set off and on June 17 had the good fortune to sight the *Meshuda*, the 46-gun flagship of the Algerian admiral Rais Hammidia. They gave chase as the Algerians raced for their home port, the 38-gun frigate USS *Constellation* leading the pursuit.

The *Constellation* succeeded in overhauling the *Meshuda* and her broadsides soon inflicted enough damage to convince Hammidia that Algiers was an impossibly distant refuge. The admiral turned his ship toward Spain, hoping to shelter in a neutral port. With the sloop *Ontario* joining the *Constitution*, however, Meshuda was soon closely engaged. Decatur then brought his flagship *Guerriere* alongside the Algerian ship and devastated it with point-blank broadsides. The Algerian admiral was killed but the crew fought on, inflicting casualties with musket fire from aloft. Finally the American sloop *Epervier* fired broadside after broadside into the Algerian ship, persuading the survivors to strike the colors.

Decatur sailed on to Algiers, whose bey soon accepted American terms, as did the rulers of Tripoli and Tunis. The following year, after the bombardment of Algiers by an Anglo-Dutch fleet, the privateering of the Barbary states was definitively brought to an end.